

SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

***CO2:** Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

sDry Run Evaluation

Code of Conduct:

I _____V. Asha Sarthiwika(192311066)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student :V. Asha Sarthiwika

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph: A

/ \

B C

/\ /\

D E F G

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration Queue Visited Node

1
2
3
4
5
6
7

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

A

/ \

B C

/\ /\

D E F G

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. Initial State :

1 3 6
5 0 2
4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion

1
2
3
4
5

Questions

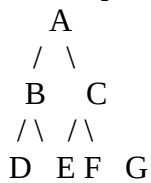
1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

Question 1: Breadth-First Search (BFS)

Given Graph



BFS uses a Queue (FIFO).

Initially

- Queue = [A]
- Visited = { }

BFS Dry Run

Iteration Queue Before Removing Visited Node Queue After Expansion

1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

BFS Traversal Order

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

Question 2

Queue after Every Iteration

Iteration Queue

Start A

After A B, C

After B C, D, E

After C D, E, F, G

After D E, F, G

After E F, G

After F G

After G Empty

Question 2: Depth-First Search (DFS)

DFS uses a Stack (LIFO).

Assume left child is explored first.

DFS Dry Run

Iteration Stack Before Pop Visited Node Stack After Expansion

1	A	A	C, B
---	---	---	------

Iteration	Stack Before Pop	Visited Node	Stack After Expansion
2	C, B	B	C, E, D
3	C, E, D	D	C, E
4	C, E	E	C
5	C	C	G, F
6	G, F	F	G
7	G	G	Empty

DFS Traversal

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Question 3: 8 Puzzle using BFS

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Important Note

This initial state is many moves away from the goal.

Therefore,

the goal cannot be reached in only 5 iterations.

A complete BFS would require expanding many more states.

For an exam dry run, we normally trace only the first few iterations.

Iteration 1

Current State

1 3 6

5 0 2

4 7 8

Blank is in the center.

Possible moves

- Up
- Down
- Left
- Right

Generated states

Move Up

1 0 6

5 3 2

4 7 8

Move Down

1 3 6

5 7 2

4 0 8

Move Left

1 3 6

0 5 2

4 7 8

Move Right

1 3 6

5 2 0

4 7 8

Queue After Expansion

S1

S2

S3

S4

Iteration 2

Expand

1 0 6

5 3 2

4 7 8

Generate new legal states except already visited.

Queue continues growing.

Iteration 3

Expand

1 3 6

5 7 2

4 0 8

Generate children.

Iteration 4

Expand

1 3 6

0 5 2

4 7 8

Generate children.

Iteration 5

Expand

1 3 6

5 2 0

4 7 8

Generate children.

Dry Run Table

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	Initial State	Initial	4 successor states
2	Move Up	Remaining 3 states	New states added
3	Move Down	Queue updated	New states added
4	Move Left	Queue updated	New states added

Iteration	Current State	Queue Before Expansion	Queue After Expansion
5	Move Right	Queue updated	Queue continues